

The Relationship Between Cantor's Theorem and Machine Learning

Eren Cice

Abstract—Cantor's Theorem states that the power set of any set (the set formed by all of its subsets) always has a strictly greater cardinality than the set itself. This paper examines the relationship between the notion of size introduced by this theorem and machine learning, through the concept of the hypothesis space — the counterpart of this notion in statistical learning theory. The concept of shattering and the Vapnik–Chervonenkis (VC) Dimension are explained through examples of one-dimensional threshold classifiers and two-dimensional linear classifiers, and the relationship between model capacity and generalization performance is discussed.

Index Terms—Cantor's Theorem, cardinality, hypothesis space, shattering, VC Dimension

I. Introduction

Cantor's Theorem states that the number of subsets of any set is always greater than the number of elements in the set itself. This result shows that, particularly in the case of infinite sets, the notion of size in mathematics cannot be explained solely by the number of elements. From the standpoint of machine learning, this idea forms an important theoretical foundation, since the number of possible classifications that can be constructed from an input space is extremely large.

II. Cantor's Theorem and the Concept of Infinity

Based on Cantor's diagonal argument, this theorem shows that the proposition “the power set is larger,” which is intuitively obvious for finite sets, also holds for infinite sets. Even the power set of a countably infinite set, such as the set of natural numbers, has a cardinality too large to be placed in one-to-one correspondence with the set itself. This finding shows that infinity is not a single concept: sets of different, mutually incomparable infinite sizes can exist.

$$|P(S)| > |S|$$

Formally, let S be any set and $P(S)$ its power set. Cantor's Theorem states that this inequality holds for every S , meaning that no surjective function from S onto $P(S)$ can exist.

III. The Hypothesis Space in Machine Learning

In machine learning, the hypothesis space refers to the set of all possible functions or models that a learning algorithm can select from the training data. Since every distinct subset of the input space can, in principle, represent a different classification pattern, the size of the hypothesis space grows extremely fast. As a natural consequence, it is not possible to learn all functions

in the hypothesis space using only a finite number of training examples.

For this reason, learning algorithms aim to improve generalization performance by restricting the hypothesis space under certain assumptions; however, the mere size of a hypothesis space does not imply that it will yield successful learning. What matters is the extent to which this space can distinguish between different data sets and represent different labeling patterns. For this reason, statistical learning theory required a measure that could quantitatively assess the learning capacity of a hypothesis space. The Vapnik–Chervonenkis Dimension (VC Dimension), developed for this purpose, is one of the fundamental theoretical concepts that measures the representational power and complexity of a hypothesis space [1], [2].

IV. The Concept of Shattering

In order to understand the VC Dimension, the concept of shattering must first be explained. Shattering means that a hypothesis space can correctly classify a given sample set for every possible labeling that can be formed on it. In other words, if for every combination of positive/negative labels assigned to the points in a sample set there exists a hypothesis that realizes that labeling, then that sample set is said to be shattered by the hypothesis space.

A. One-Dimensional Threshold Classifiers

One-dimensional threshold classifiers are among the simplest examples of this concept. While all possible labelings can be realized for a single sample, not all possible labeling combinations can be obtained once two samples are involved. This shows that no hypothesis space has unlimited discriminative capacity, and that shattering ability remains bounded within certain limits.

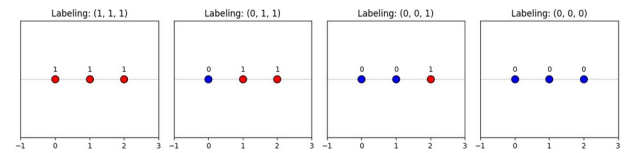


Fig. 1. Shattering example with a one-dimensional threshold classifier.

B. Two-Dimensional Linear Classifiers

As the capacity of a hypothesis space increases, so does the number of samples it can shatter. For example, in two-dimensional linear classifiers, all possible labeling

combinations can be realized for any three points that do not lie on the same line. Linear classifiers can therefore fully shatter three samples.

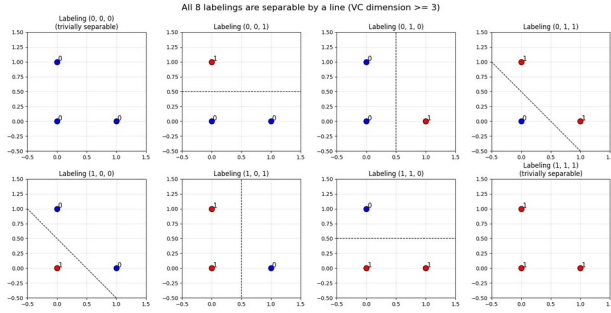


Fig. 2. Shattering of three points by a linear classifier across all labeling combinations.

C. The Four-Point Case and the Limits of Shattering

However, the situation changes when the number of samples is four. Although there are sixteen distinct labelings for four points, not all of them can be realized with a single line. In other words, linear classifiers cannot fully shatter four samples. For this reason, the VC Dimension of linear classifiers is taken to be 3.

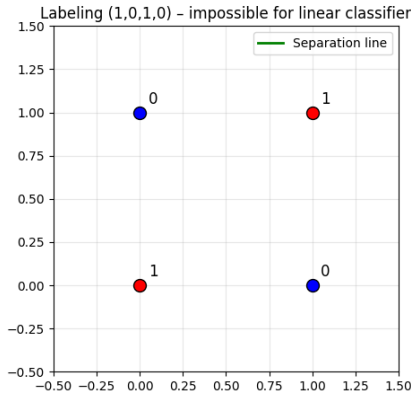


Fig. 3. A labeling of four points (XOR-like) that cannot be shattered by a linear classifier.

V. VC Dimension and Model Capacity

These examples show that the capacity of a hypothesis space cannot be explained solely by the number of hypotheses it contains, but also depends on its ability to separate different sample sets in all possible ways. The VC Dimension is the fundamental measure that quantifies this capacity. Insufficient model capacity leads to underfitting, while excessive capacity leads to overfitting. For this reason, a reliable learning process requires maintaining a balance not only in the size of the hypothesis space, but also between the number of samples (n) and the VC Dimension (d) [3].

VI. Conclusion

The theoretical notion of size introduced by Cantor's Theorem provides a powerful framework for conceptualizing

the complexity of the hypothesis space in machine learning. The VC Dimension, in turn, serves as the fundamental measure that quantitatively evaluates the learnable portion of that complexity. In conclusion, Cantor's Theorem — a purely theoretical result concerning the cardinality of infinite sets — provides a conceptual foundation for understanding the balance between model capacity and generalization performance in statistical learning theory.

References

- [1] V. N. Vapnik and A. Y. Chervonenkis, "On the uniform convergence of relative frequencies of events to their probabilities," *Theory of Probability & Its Applications*, vol. 16, no. 2, pp. 264–280, 1971.
- [2] V. N. Vapnik, *Statistical Learning Theory*. New York, NY, USA: Wiley, 1998.
- [3] S. Shalev-Shwartz and S. Ben-David, *Understanding Machine Learning: From Theory to Algorithms*. Cambridge, U.K.: Cambridge Univ. Press, 2014.